



NCCER Module 22103



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Identification of Heavy Equipment





Section Three

Hydraulic System

Objective

3. Explain hydraulic systems and their components.
 - a. Explain hydraulic system fundamentals.
 - b. Identify and describe the equipment and components used in hydraulic systems.



Performance Task

Under the supervision of an instructor, trainees should be able to do the following:

There are no Performance Tasks associated with this section.



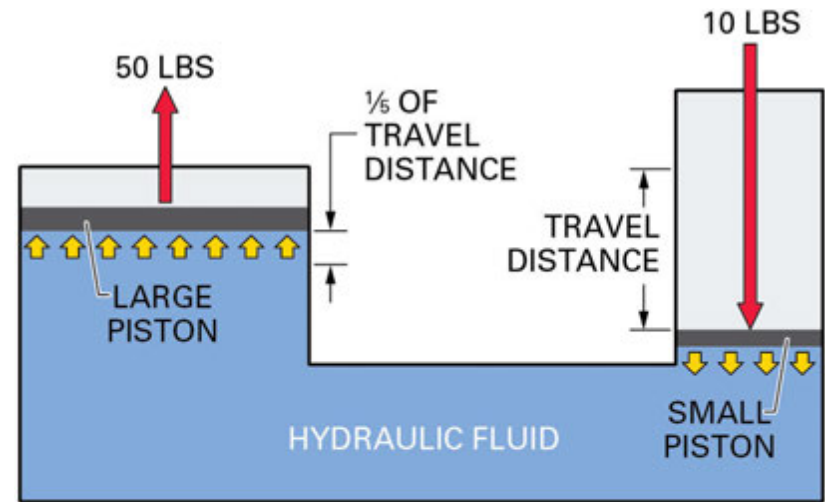
3.0.0 Hydraulic System

- The hydraulic system is used to control the positioning of the blades, buckets, and other tools used on most heavy construction equipment.
- A hydraulic system uses hydraulic pressure on hydraulic fluids to do its work.



3.1.0 Hydraulic Fundamentals (1 of 2)

Pascal's law of hydraulics states that any pressure applied to a fluid in a closed system is applied to all points and surfaces in the system equally.



FORMULAS FOR PASCAL'S LAW

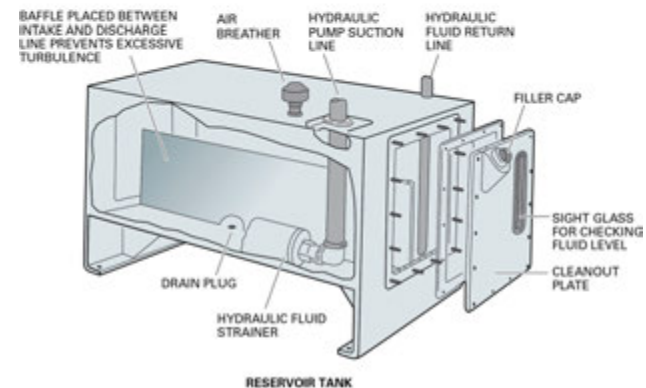
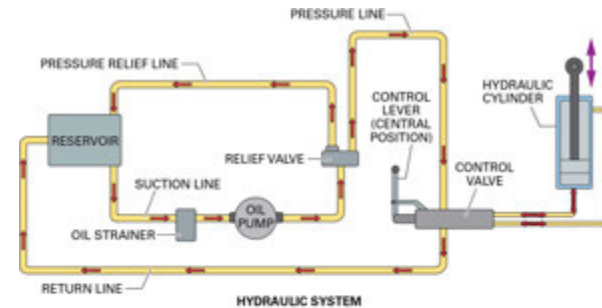
$$\text{PRESSURE (PSI)} = \frac{\text{FORCE (LBS)}}{\text{AREA (IN}^2\text{)}}$$

$$\text{FORCE (LBS)} = \text{PRESSURE (PSI)} \times \text{AREA (IN}^2\text{)}$$



3.1.0 Hydraulic Fundamentals (2 of 2)

- A typical hydraulic system consists of a hydraulic pump, a reservoir with a strainer, a valve, and a **hydraulic actuator**, in this case a cylinder.
- Hydraulics increase a system's mechanical advantage.
- Hydraulic oil may be very hot if the machine has been operating.



3.2.0 Hydraulic Equipment and Components

Hydraulic systems are made up of components and hydraulic hoses. The main components of hydraulic systems used in heavy equipment are brakes, clutches, couplers, and pumps.



3.2.1 Hydraulic Brakes and Clutches

- Power-assist brakes use a small reservoir of brake fluid with a piston. As the operating pedal is pushed, it transmits the pressure to a brake cylinder, which in turn applies the brake pad to the drum or disc.
- The clutch system involves a pedal operating a piston in a reservoir. The pressure applied when the pedal is pushed is transmitted to a cylinder that moves to overcome the spring pressure of the pressure plate. This allows the engine power to be disconnected from the power distribution system.



3.2.2 Hydraulic Power Couplers

- A **torque converter** is a small, self-contained hydraulic system that connects the main power source to the transmission or gearbox.
- It replaces a conventional clutch system.
- The pump forces fluid into the turbine vanes, which causes the transmission gearing to transfer the torque to the power distribution system.



3.2.3 Hydraulic Pumps

- The pump is driven from the output of the transmission or gearbox to supply the fluid flow necessary for operation of the hydraulic power plant.
- If the engine speed is kept too low, the hydraulic system will have insufficient flow to operate the system.
- Reduced flow can cause overheating of the hydraulic oil and prevent the operation of some or all of the hydraulic actuators.
- The equipment operator must maintain enough engine speed to allow for proper operation of the hydraulic system.



Wrap Up – Trade Terms

Dogged: Held or fastened tightly with a mechanical device.

Hydraulic actuator: A device that converts the fluid energy of the hydraulic system into mechanical energy using a hydraulic cylinder or motor.

Pascal's law: A scientific law stating that pressure at any one point in a closed static fluid system will be applied to all points in the system in every direction and will exert equal forces on equal areas. It is the basis of hydraulic theory and was developed by Blaise Pascal (1623–1662), a French philosopher and mathematician.

Torque converter: A device that uses fluid, usually oil, to transmit torque from one shaft to another. Also known as a fluid coupler.



3.0.0 Section Review Question 1

1. The hazards associated with hydraulic systems are _____.
- a. heat and high pressure
 - b. high pressure and toxic fumes
 - c. heat and toxic fumes
 - d. heat and corrosive acid



3.0.0 Section Review Question 1 Answer

1. The hazards associated with hydraulic systems are _____.

- a. heat and high pressure**
- b. high pressure and toxic fumes
- c. heat and toxic fumes
- d. heat and corrosive acid



3.0.0 Section Review Question 2

2. The device used to connect the engine to the transmission in place of a clutch is the ____.
- a. hydraulic actuator
 - b. hydrostatic shifter
 - c. pressure regulator
 - d. torque converter



3.0.0 Section Review Question 2 Answer

2. The device used to connect the engine to the transmission in place of a clutch is the ____.
- a. hydraulic actuator
 - b. hydrostatic shifter
 - c. pressure regulator
 - d. torque converter**



Next...



Module Review and Examination

Complete the Review Questions at the end of the module and prepare for the Module Exam.